

SI for *Museum specimens* reveal butterfly size
increases through time but divergent responses to
temperature

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1 SI Tables

Table S1: Details of butterfly species used in this study. Habitat and foodplant specialism status was split into two categories divided specialists (S) and generalists (G). Numbers for each datasets are the post-filtering counts. See code repository for corresponding .csv file.

<i>Species</i>	Common Name	Family	Habitat	Foodplant	n (NHM)	n (OUMNH)	Focal Brood	Focal Month (Late Larval)	GB Red List category
<i>Aglaia urticae</i>	Small tortoiseshell	Nymphalidae	G	S	971	34	1: Jun-Aug	May	Not Listed
<i>Aricia agestis</i>	Brown argus	Lycaenidae	S	S	975	42	2:Aug-Oct	July	Not Listed
<i>Celastrina argiolus</i>	Holly blue	Lycaenidae	G	G	303	25	2:Aug-Oct	June	Not Listed
<i>Lasiommata megera</i>	Wall	Nymphalidae	G	G	619	62	2: Aug-Oct	June	Endangered
<i>Leptidea sinapis</i>	Wood White	Pieridae	S	S	791	70	1: Mar-Jun	JulyBefore	Endangered
<i>Lycaena phlaeas</i>	Small copper	Lycaenidae	G	S	2877	65	2:Jul-Oct	June	Not Listed
<i>Pararge aegeria</i>	Speckled wood	Nymphalidae	G	G	471	38	2:Jul-Aug	June	Not Listed
<i>Pieris brassicae</i>	Large white	Pieridae	G	S	344	44	2:Jul-Sep	June	Not Listed
<i>Pieris napi</i>	Green-veined white	Pieridae	G	G	766	NA	2:Jul-Sep	June	Not Listed
<i>Pieris rapae</i>	Cabbage white	Pieridae	G	G	445	NA	2: Jul-Sep	June	Not Listed
<i>Polygonia c-album</i>	Comma	Nymphalidae	G	G	466	39	1: Jun-Aug	June	Not Listed
<i>Polyommatus bellargus</i>	Adonis Blue	Lycaenidae	S	S	4830	63	2: Aug-Sep	July	Vulnerable
<i>Polyommatus icarus</i>	Common blue	Lycaenidae	G	G	3752	63	2:Aug-Sep	July	Not Listed
<i>Aglaia io</i>	Peacock	Nymphalidae	G	S	505	80	Univoltine	June	Not Listed
<i>Anthocharis car-damines</i>	Orange tip	Pieridae	G	G	1661	64	Univoltine	JulyBefore	Not Listed
<i>Apatura iris</i>	Purple emperor	Nymphalidae	S	S	124	NA	Univoltine	March	Not Listed
<i>Aphantopus hyperan-tus</i>	Ringlet	Nymphalidae	G	G	2206	NA	Univoltine	June	Not Listed
<i>Argynnis paphia</i>	Silver washed fritillary	Nymphalidae	G	S	1447	83	Univoltine	June	Not Listed

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Table S1 – *Continued from previous page*

<i>Species</i>	Common Name	Family	Habitat	Foodplant	n (NHM)	n (OUMNH)	Focal Brood	Focal Month (Late Larval)	GB Red List category
<i>Boloria euphrosyne</i>	Pearl-bordered Fritillary	Nymphalidae	G	S	1865	NA	Univoltine	April	Vulnerable
<i>Boloria selene</i>	Small Pearl-bordered Fritillary	Nymphalidae	S	S	1925	NA	Univoltine	April	Vulnerable
<i>Callophrys rubi</i>	Green hairstreak	Lycaenidae	G	G	990	NA	Univoltine	JuneBefore	Not Listed
<i>Carterocephalus palaemon</i>	Chequered Skipper	Hesperiidae	S	S	604	NA	Univoltine	AugustBefore	Not Listed
<i>Coenonympha pamphilus</i>	Small Heath	Nymphalidae	G	S	3058	NA	Univoltine	April	Vulnerable
<i>Coenonympha tullia</i>	Large Heath	Nymphalidae	S	S	1409	44	Univoltine	April	Endangered
<i>Cupido minimus</i>	Small Blue	Lycaenidae	G	S	1453	89	Univoltine	JulyBefore	Near Threatened
<i>Erynnis tages</i>	Dingy skipper	Hesperiidae	G	S	856	85	Univoltine	AugustBefore	Not Listed
<i>Euphydryas aurinia</i>	Marsh Fritillary	Nymphalidae	G	S	1926	NA	Univoltine	May	Vulnerable
<i>Fabriciana adippe</i>	High Brown Fritillary	Nymphalidae	G	S	994	NA	Univoltine	May	Endangered
<i>Favonius quercus</i>	Purple hairstreak	Lycaenidae	G	S	494	83	Univoltine	May	Not Listed
<i>Gonepteryx rhamni</i>	Brimstone	Pieridae	G	S	483	81	Univoltine	June	Not Listed
<i>Hamearis lucina</i>	Duke of Burgundy	Riodinidae	S	S	951	NA	Univoltine	JulyBefore	Vulnerable
<i>Hesperia comma</i>	Silver-spotted Skipper	Hesperiidae	S	S	646	103	Univoltine	June	Vulnerable
<i>Hipparchia semele</i>	Grayling	Nymphalidae	S	G	1654	NA	Univoltine	April	Endangered
<i>Limenitis camilla</i>	White Admiral	Nymphalidae	S	S	761	97	Univoltine	May	Vulnerable
<i>Maniola jurtina</i>	Meadow brown	Nymphalidae	G	G	4730	70	Univoltine	April	Not Listed
<i>Melanargia galathea</i>	Marbled white	Nymphalidae	G	G	2169	101	Univoltine	May	Not Listed
<i>Melitaea athalia</i>	Heath Fritillary	Nymphalidae	G	G	1619	NA	Univoltine	March	Endangered
<i>Melitaea cinxia</i>	Glanville Fritillary	Nymphalidae	S	S	579	NA	Univoltine	March	Endangered
<i>Nymphalis polychloros</i>	Large Tortoiseshell	Nymphalidae	G	G	197	NA	Univoltine	June	Regionally Extinct

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Table S1 – *Continued from previous page*

<i>Species</i>	Common Name	Family	Habitat	Foodplant	n (NHM)	n (OUMNH)	Focal Brood	Focal Month (Late Larval)	GB Red List category
<i>Ochlodes sylvanus</i>	Large skipper	Hesperiidae	G	G	1029	NA	Univoltine	May	Not Listed
<i>Papilio machaon</i>	Swallowtail	Papilionidae	S	S	166	54	Univoltine	JulyBefore	Vulnerable
<i>Plebejus argus</i>	Silver-studded Blue	Lycaenidae	S	G	7105	NA	Univoltine	May	Vulnerable
<i>Polyommatus coridon</i>	Chalk Hill Blue	Lycaenidae	G	S	19584	NA	Univoltine	June	Vulnerable
<i>Pyrgus malvae</i>	Grizzled Skipper	Hesperiidae	G	G	1285	108	Univoltine	July	Vulnerable
<i>Pyronia tithonus</i>	Gatekeeper	Nymphalidae	G	S	2313	125	Univoltine	June	Not Listed
<i>Satyrrium pruni</i>	Black Hairstreak	Lycaenidae	S	S	377	NA	Univoltine	May	Endangered
<i>Satyrrium w-album</i>	White-letter Hairstreak	Lycaenidae	S	S	346	NA	Univoltine	May	Vulnerable
<i>Speyeria aglaja</i>	Dark Green Fritillary	Nymphalidae	G	S	1028	51	Univoltine	May	Near Threat- ened
<i>Thecla betulae</i>	Brown Hairstreak	Lycaenidae	G	S	165	NA	Univoltine	June	Vulnerable
<i>Thymelicus acteon</i>	Lulworth Skipper	Hesperiidae	S	S	544	NA	Univoltine	May	Near Threat- ened
<i>Thymelicus lineola</i>	Essex skipper	Hesperiidae	G	G	707	NA	Univoltine	May	Not Listed
<i>Thymelicus sylvestris</i>	Small skipper	Hesperiidae	G	G	918	98	Univoltine	May	Not Listed

Table S2: Data underlying Figure 4b. Fitted temperature - reponse coefficients, grouped by UK Butterfly Red List status.

<i>Species</i>	RedListing	GB Red List category	mean	q5	q95
<i>Thecla betulae</i>	Listed	Vulnerable	-0.0080	-0.0148	-0.0011
<i>Nymphalis polychloros</i>	Listed	Regionally Extinct	-0.0066	-0.0177	0.0046
<i>Leptidea sinapis</i>	Listed	Endangered	-0.0045	-0.0072	-0.0020
<i>Coenonympha tullia</i>	Listed	Endangered	-0.0038	-0.0058	-0.0017
<i>Pyrgus malvae</i>	Listed	Vulnerable	-0.0033	-0.0052	-0.0012
<i>Hamearis lucina</i>	Listed	Vulnerable	-0.0019	-0.0040	0.0001
<i>Boloria selene</i>	Listed	Vulnerable	-0.0010	-0.0034	0.0013
<i>Polyommatus bellargus</i>	Listed	Vulnerable	-0.0009	-0.0018	0.0001
<i>Cupido minimus</i>	Listed	Near Threatened	-0.0005	-0.0033	0.0023
<i>Satyrrium pruni</i>	Listed	Endangered	-0.0002	-0.0035	0.0030
<i>Polyommatus coridon</i>	Listed	Vulnerable	0.0005	-0.0003	0.0014
<i>Coenonympha pamphilus</i>	Listed	Vulnerable	0.0009	-0.0009	0.0028
<i>Boloria euphrosyne</i>	Listed	Vulnerable	0.0011	-0.0007	0.0028
<i>Lasiommata megera</i>	Listed	Endangered	0.0011	-0.0023	0.0045
<i>Plebejus argus</i>	Listed	Vulnerable	0.0013	-0.0003	0.0029
<i>Papilio machaon</i>	Listed	Vulnerable	0.0018	-0.0061	0.0098
<i>Satyrrium w-album</i>	Listed	Vulnerable	0.0029	-0.0024	0.0081
<i>Hipparchia semele</i>	Listed	Endangered	0.0030	0.0001	0.0059
<i>Melitaea athalia</i>	Listed	Endangered	0.0040	0.0019	0.0062
<i>Limenitis camilla</i>	Listed	Vulnerable	0.0048	0.0017	0.0078
<i>Melitaea cinxia</i>	Listed	Endangered	0.0049	0.0009	0.0091
<i>Euphydryas aurinia</i>	Listed	Vulnerable	0.0050	0.0014	0.0085
<i>Speyeria aglaja</i>	Listed	Near Threatened	0.0051	0.0024	0.0077
<i>Thymelicus acteon</i>	Listed	Near Threatened	0.0076	0.0029	0.0123
<i>Hesperia comma</i>	Listed	Vulnerable	0.0088	0.0054	0.0122
<i>Fabriciana adippe</i>	Listed	Endangered	0.0108	0.0083	0.0134
<i>Apatura iris</i>	Unlisted	NA	-0.0141	-0.0207	-0.0076
<i>Celastrina argiolus</i>	Unlisted	NA	-0.0103	-0.0179	-0.0028
<i>Polygonia c-album</i>	Unlisted	NA	-0.0059	-0.0099	-0.0018
<i>Pararge aegeria</i>	Unlisted	NA	-0.0049	-0.0087	-0.0012
<i>Pieris rapae</i>	Unlisted	NA	-0.0045	-0.0086	-0.0004
<i>Argynnis paphia</i>	Unlisted	NA	-0.0041	-0.0066	-0.0017
<i>Thymelicus lineola</i>	Unlisted	NA	-0.0035	-0.0070	0.0000
<i>Gonepteryx rhamni</i>	Unlisted	NA	-0.0034	-0.0065	-0.0003
<i>Aricia agestis</i>	Unlisted	NA	-0.0031	-0.0061	-0.0002
<i>Callophrys rubi</i>	Unlisted	NA	-0.0030	-0.0063	0.0003
<i>Aphantopus hyperantus</i>	Unlisted	NA	-0.0028	-0.0051	-0.0005
<i>Anthocharis cardamines</i>	Unlisted	NA	-0.0020	-0.0045	0.0006
<i>Pieris napi</i>	Unlisted	NA	-0.0013	-0.0042	0.0016
<i>Lycaena phlaeas</i>	Unlisted	NA	-0.0010	-0.0033	0.0013
<i>Melanargia galathea</i>	Unlisted	NA	-0.0008	-0.0037	0.0022
<i>Aglaia urticae</i>	Unlisted	NA	-0.0006	-0.0047	0.0034
<i>Maniola jurtina</i>	Unlisted	NA	0.0005	-0.0011	0.0022
<i>Thymelicus sylvestris</i>	Unlisted	NA	0.0006	-0.0019	0.0031
<i>Carterocephalus palaemon</i>	Unlisted	NA	0.0008	-0.0022	0.0038
<i>Pyronia tithonus</i>	Unlisted	NA	0.0010	-0.0009	0.0029
<i>Polyommatus icarus</i>	Unlisted	NA	0.0015	0.0000	0.0030
<i>Favonius quercus</i>	Unlisted	NA	0.0018	-0.0026	0.0063
<i>Aglaia io</i>	Unlisted	NA	0.0027	-0.0041	0.0095

Table S2: Data underlying Figure 4b. Fitted temperature - response coefficients, grouped by UK Butterfly Red List status.

<i>Species</i>	RedListing	GB Red List category	mean	q5	q95
<i>Erynnis tages</i>	Unlisted	NA	0.0033	0.0002	0.0064
<i>Ochlodes sylvanus</i>	Unlisted	NA	0.0037	0.0011	0.0064
<i>Pieris brassicae</i>	Unlisted	NA	0.0043	0.0001	0.0086

2 SI Figures

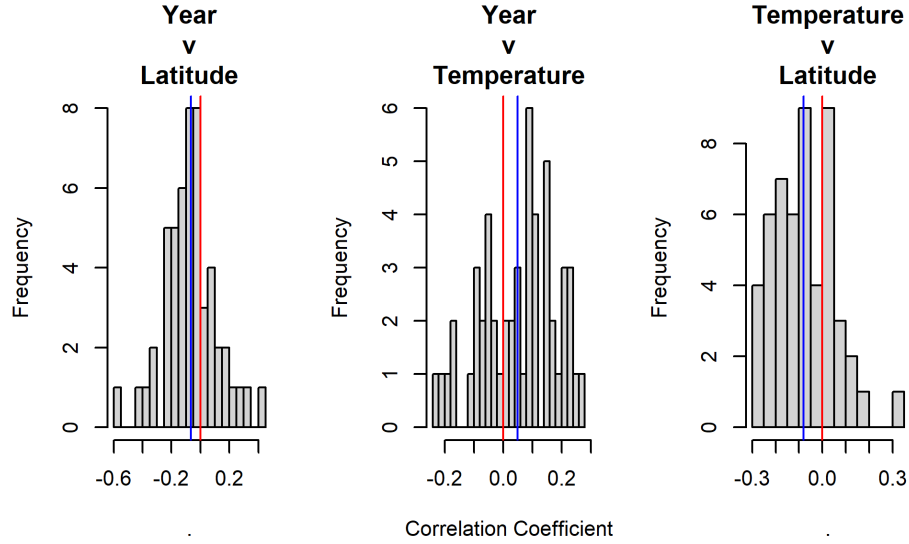


Figure S1: Distribution of within species-level correlations between key predictors in the records in the NHM dataset. Red line shows 0, blue line shows median. There is no consistent geographic trend through time. Temperature tended to be positively correlated with year, but the trend is very noisy. Likewise, there is an overall negative correlation between temperature and latitude, but is neither strong or consistent.

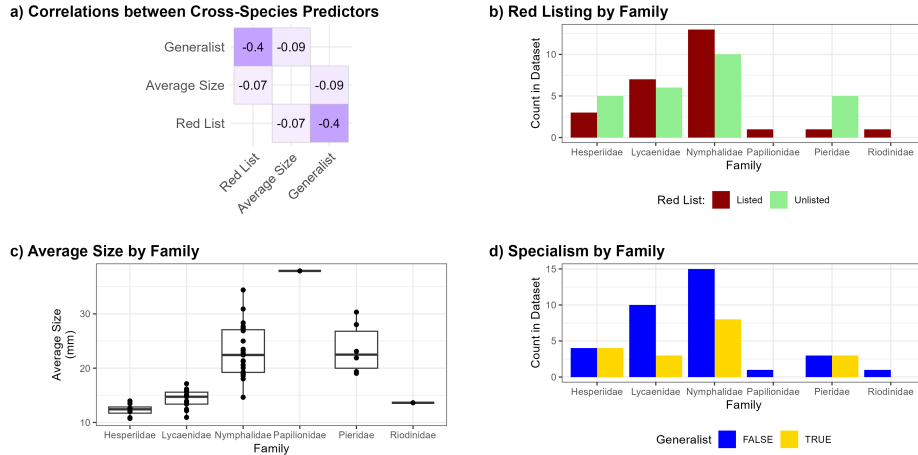


Figure S2: Associations between cross-species predictors. (A) Correlations among the three principal predictors. (B) - (D) Relationships between taxonomic family and other species-level predictors. Families vary substantially in average size, but are otherwise approximately evenly distributed across predictor space.

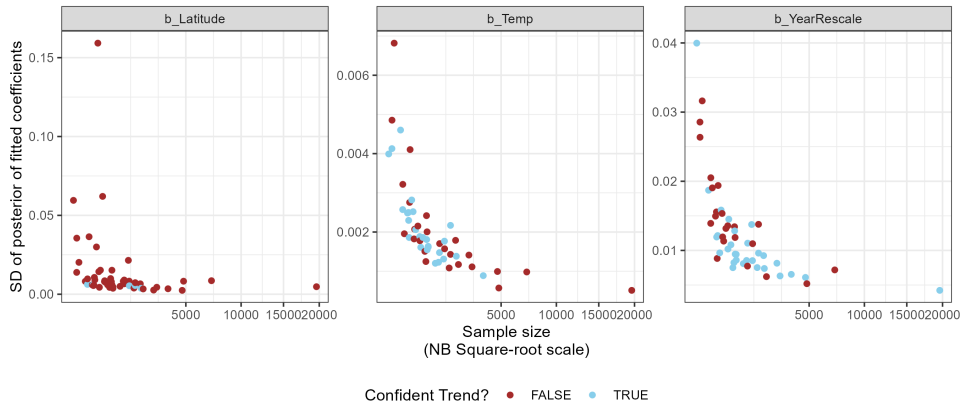


Figure S3: Relationship between sample size, confidence of posterior estimate and identification of species-level trends in the NHM dataset. Precision of posterior estimates was strongly related to the sample size for each species, but the identification of confident trends was less dependent on sample size.

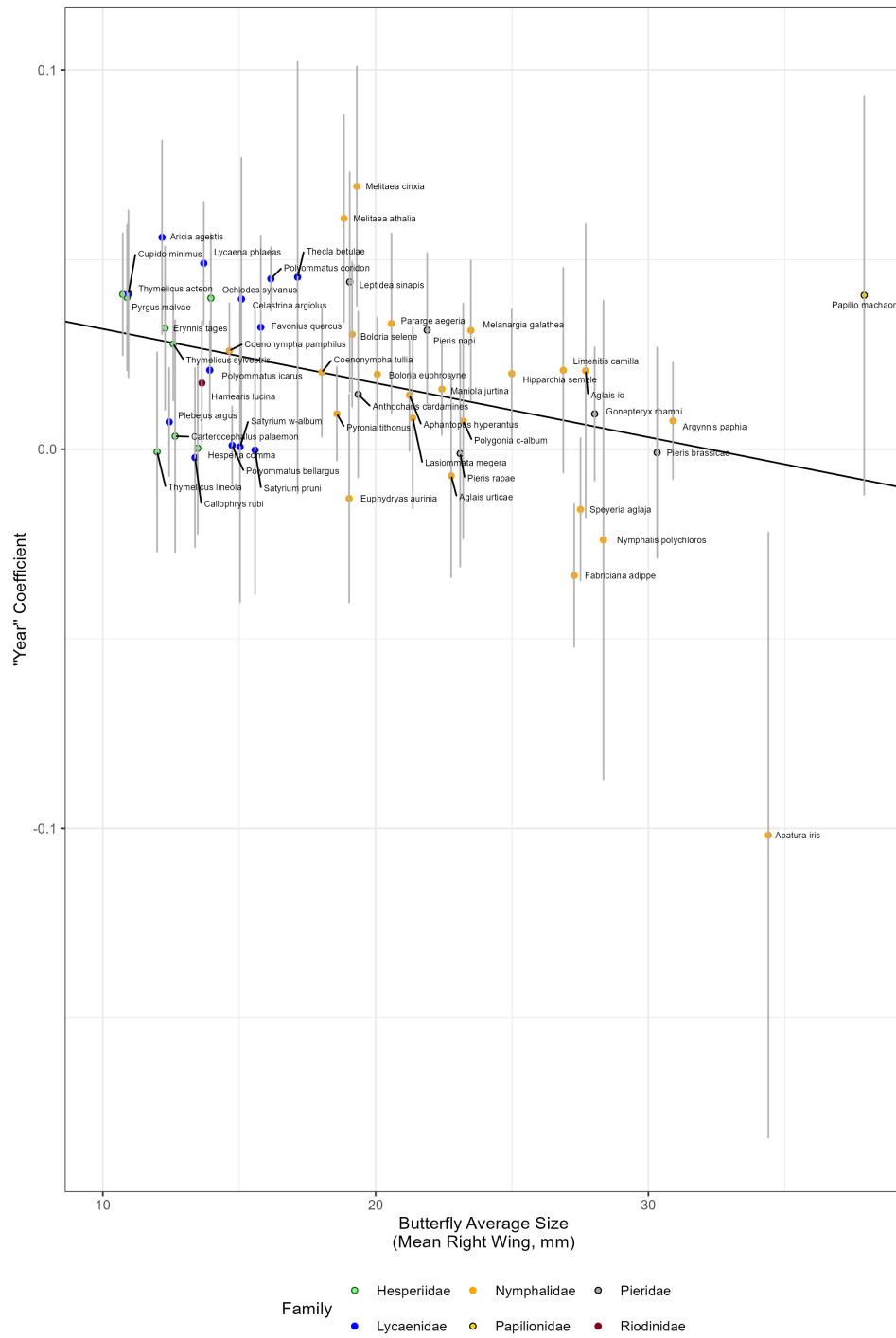
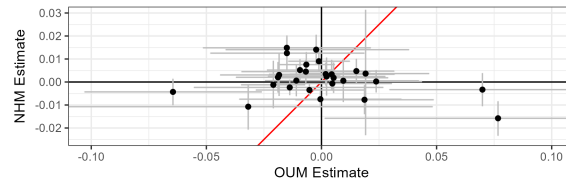
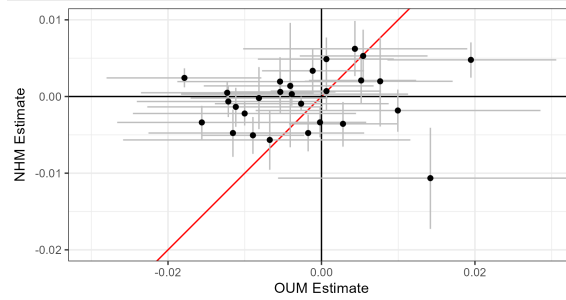


Figure S4: Main text Figure 4a with species labels and relative uncertainty of each species Error bars show $\pm$ two standard deviations of the posterior.

a) Latitude Coefficient



b) Temperature Coefficient



c) Year Coefficient

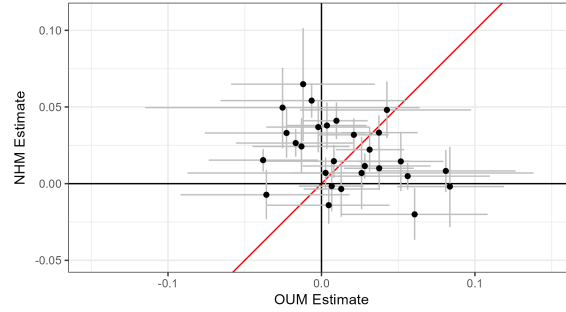


Figure S5: Coefficient Correlations between datasets. Here, each dot represents a species best-fit coefficient of the record-predictors within the two datasets. Error bars show 90% posterior confidence interval. Estimates from the smaller dataset were much more uncertain (note the unequal axis scales). Pearson's correlation of the mean estimates between the datasets was insignificant in each case (p-values: 0.158, 0.429, 0.0797 respectively (unadjusted for multiple tests)).

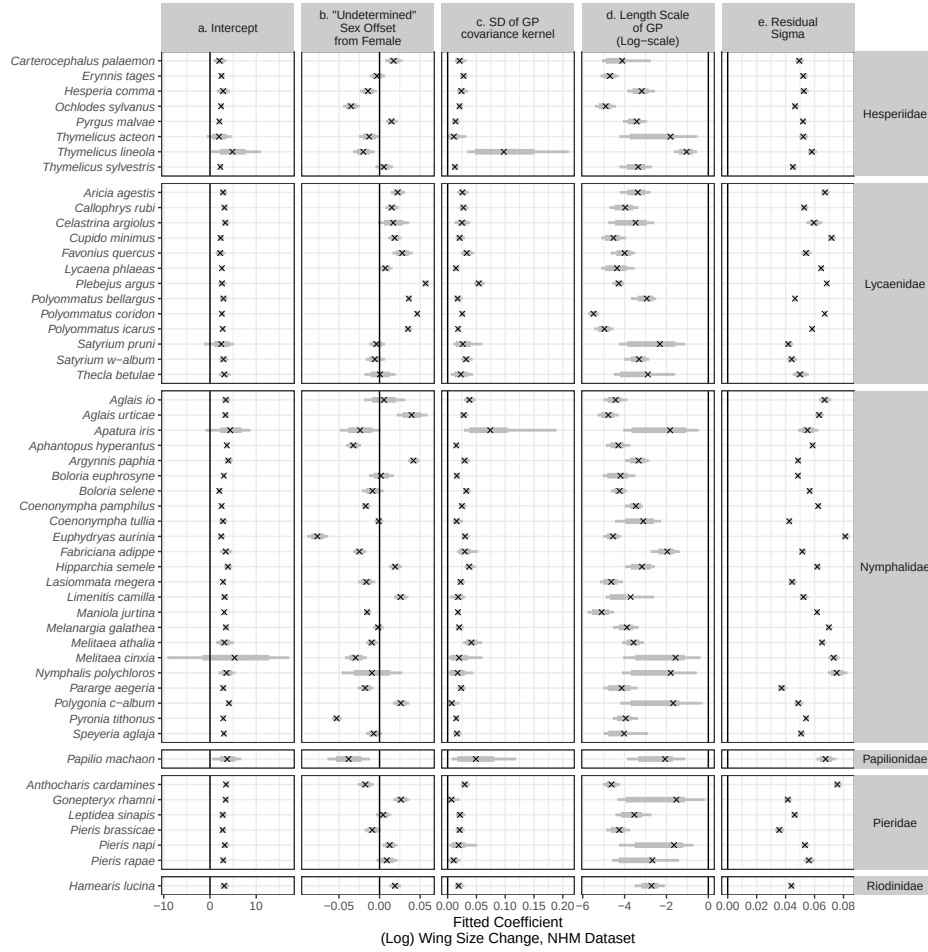


Figure S6: Non-focal fitted coefficient posterior distributions from the species level models Posterior distributions for each species for a) the overall intercept, b) the offset from the female baseline for sex-undetermined specimens. This would include a mixture of males and females, but in variable proportions, c) the standard deviation of the Gaussian process covariance kernel (logged for comparability between species), d) length-scale of the Gaussian process covariance kernel, e) estimated residual variation (σ).

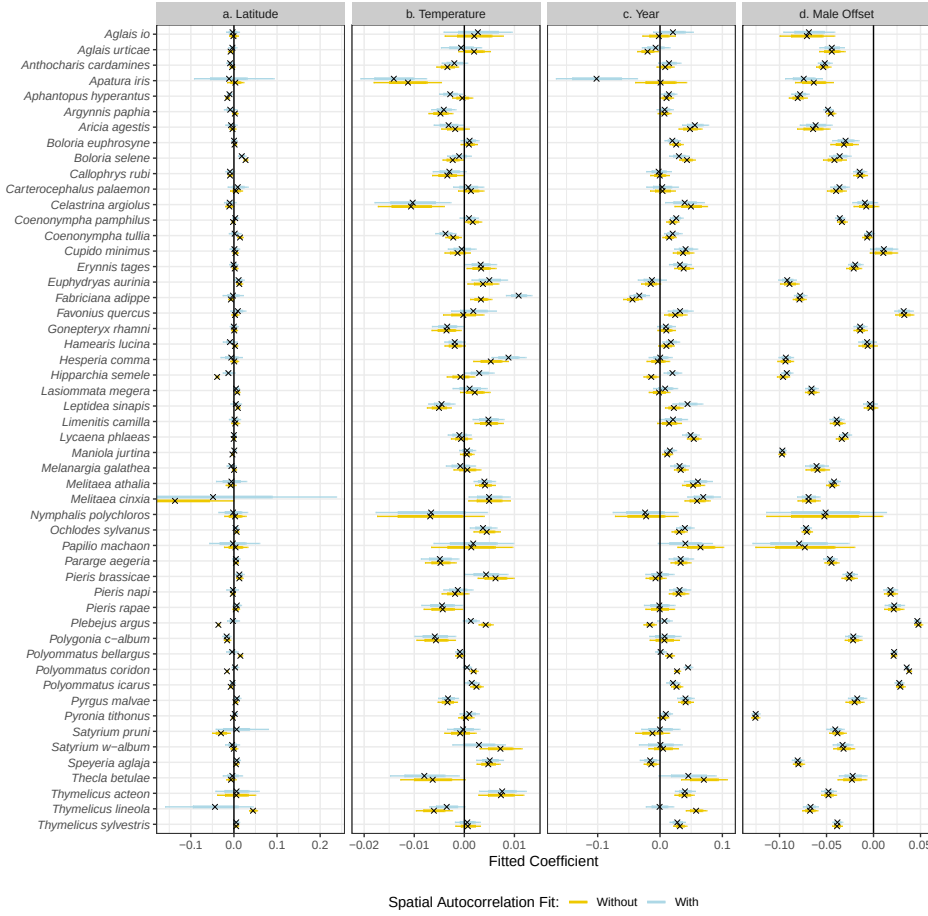


Figure S7: Impact on fitted coefficient posteriors for NHM dataset of also including spatial autocorrelation term thorough a Gaussian point process. Lines show 90% and 66% intervals. For almost all species, even though the GPP was informative for the purposes of prediction, the coefficients to be carried forward to the cross-species models are highly comparable. Cross-species regression analyses gave very similar results without the GPP term: effect of Size on Year coefficient (i.e. result presented in main text Fig 4a): -0.001435381 (95% interval: -0.002451985 to -4.021077e-04) and effect of Red Listing on Temperature coefficient (i.e. result presented in main text Fig 4b): -0.00217 (95% interval:-0.0041 to -0.00018).

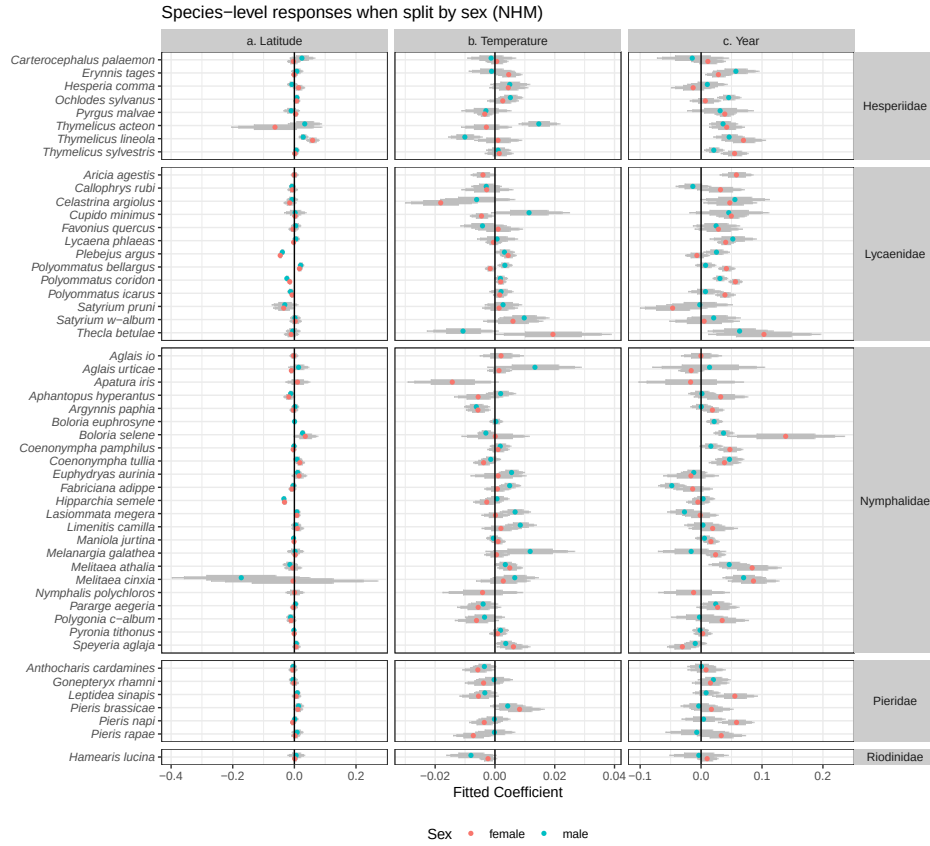


Figure S8: Investigation of sex-specific responses to predictor variables Species-level regressions models were refit (without the spatial term) for each sex separately to investigate possible sex-specific responses to the predictors. Note that the degree and balance of sex-determination varied considerably between species due to ease of identification - for five species (*Aglaia io*, *Apatura iris*, *Aricia agestis*, *Nymphalis polychloros*, *Boloria euphrosyne*) these splits were not possible. Grey lines show 95%, 90% and 66% intervals. Direct comparison of posteriors identified that in 17/138 comparisons the central 95% of the posterior difference did not overlap zero.

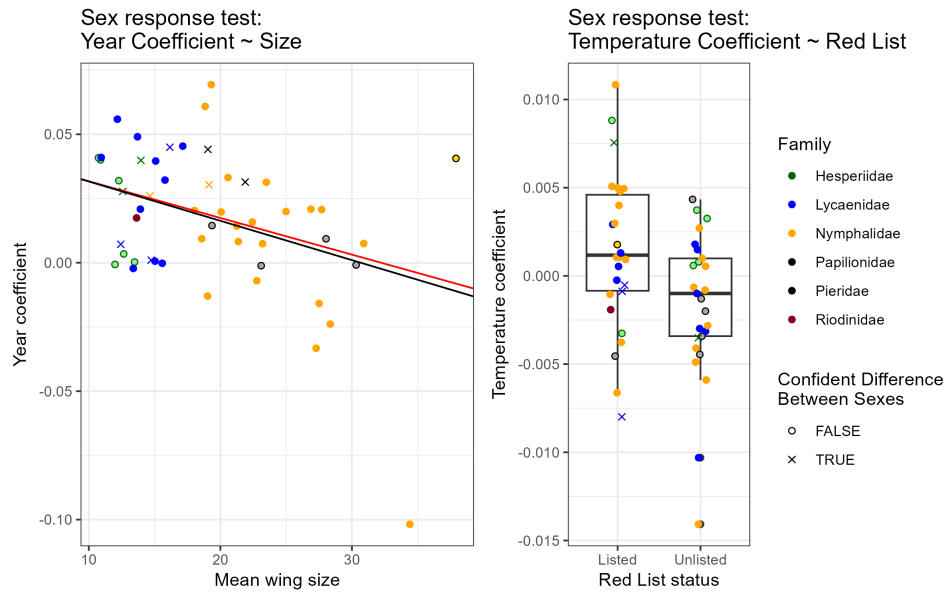
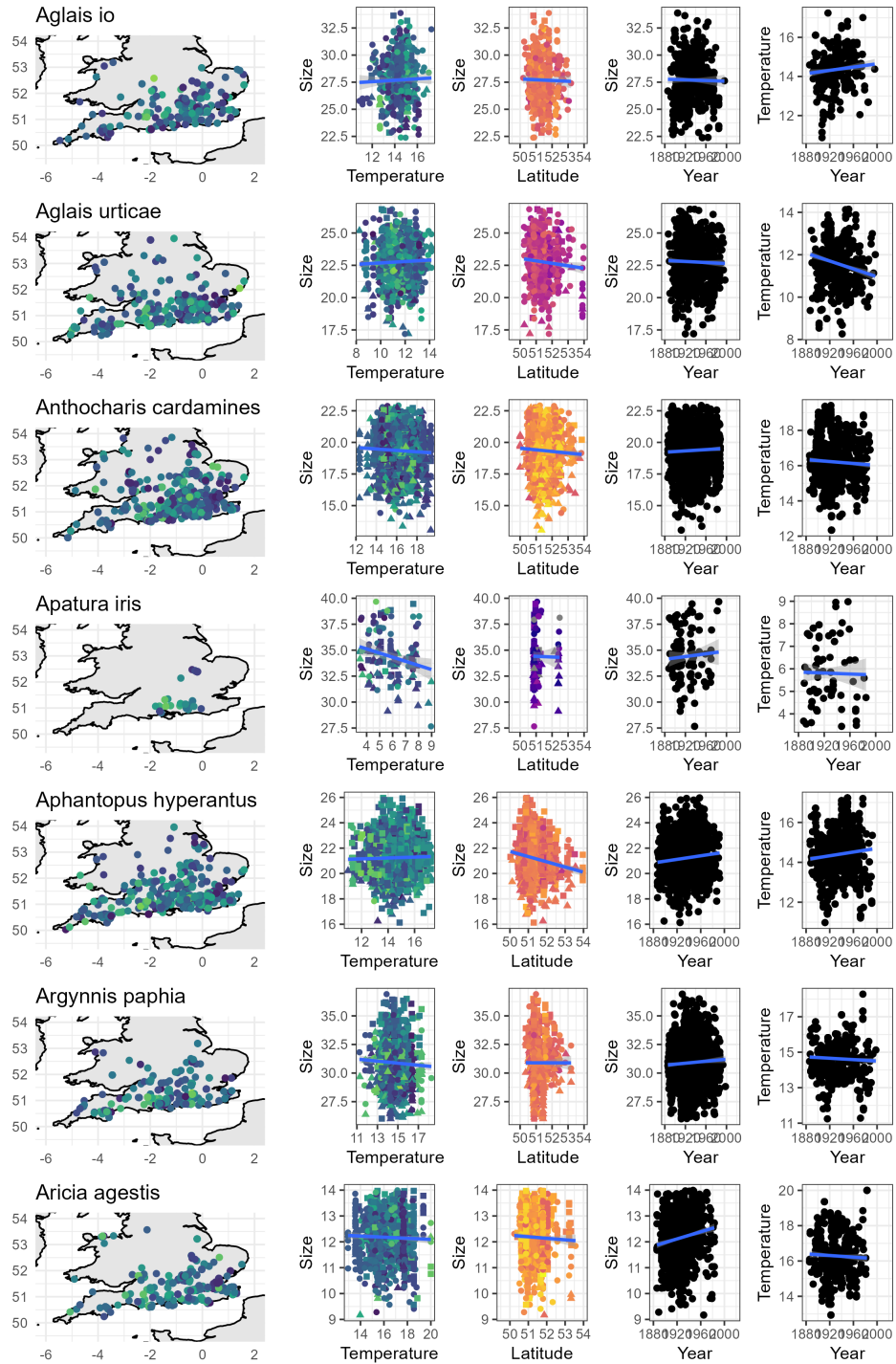
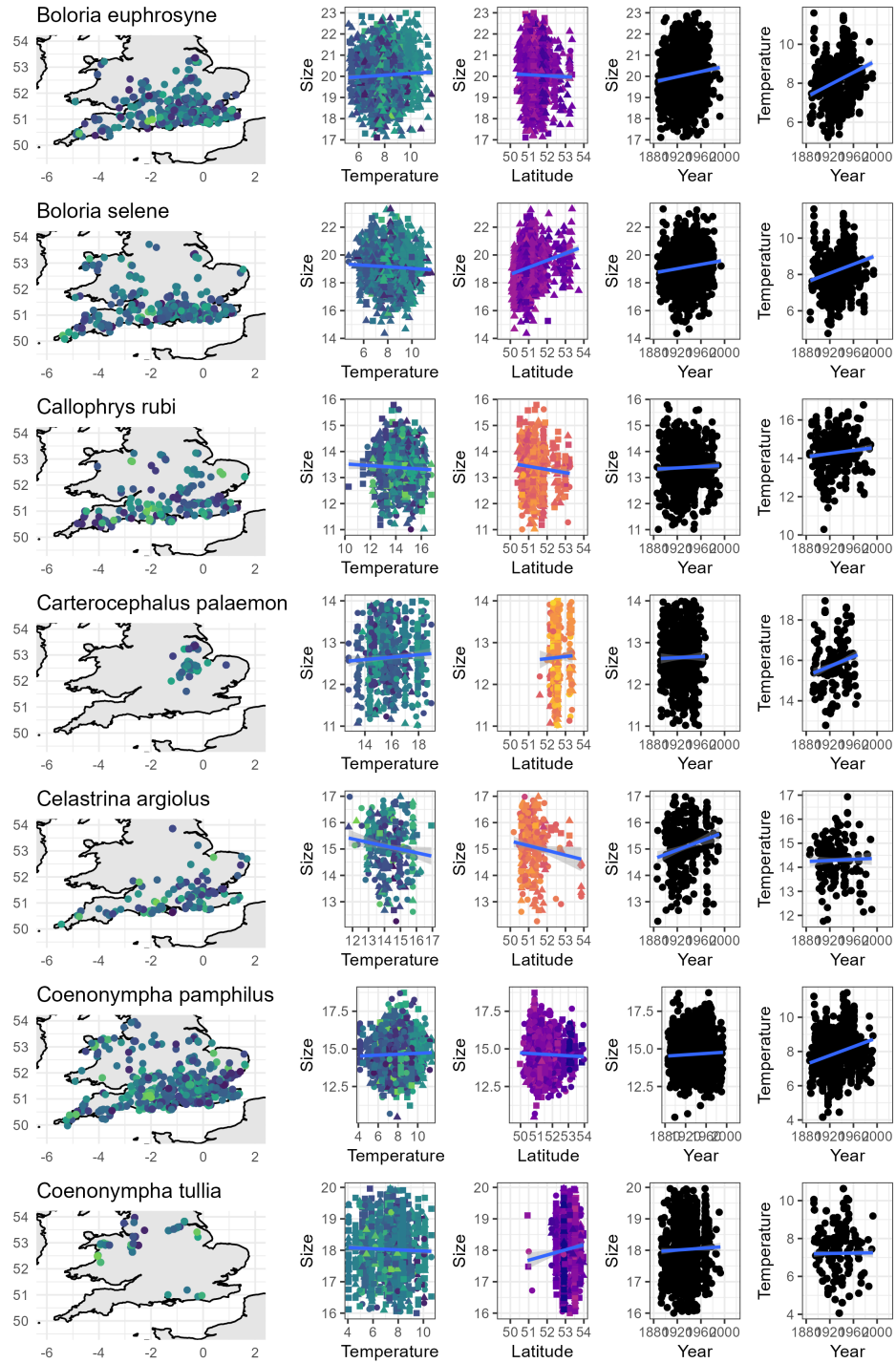
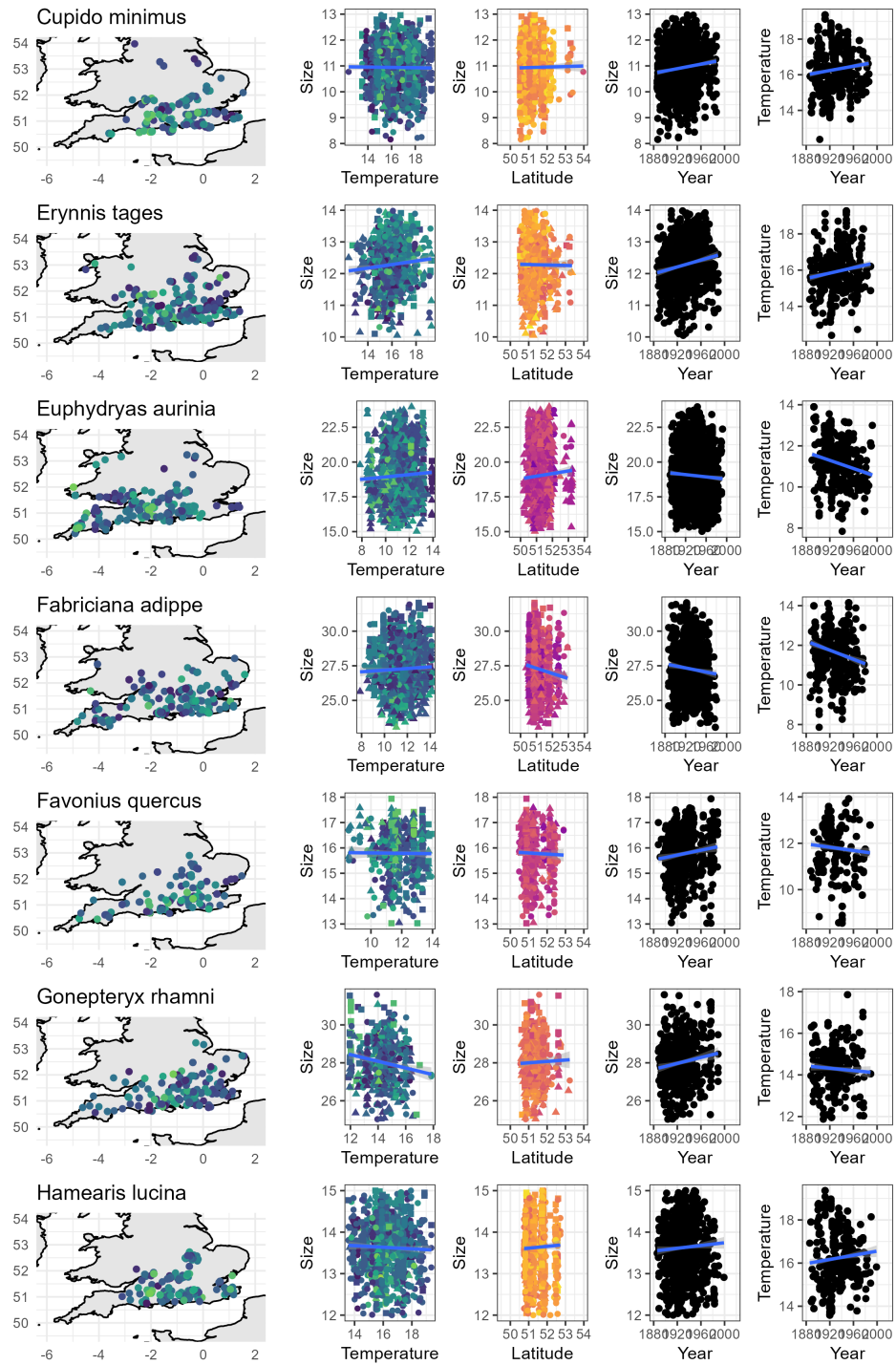


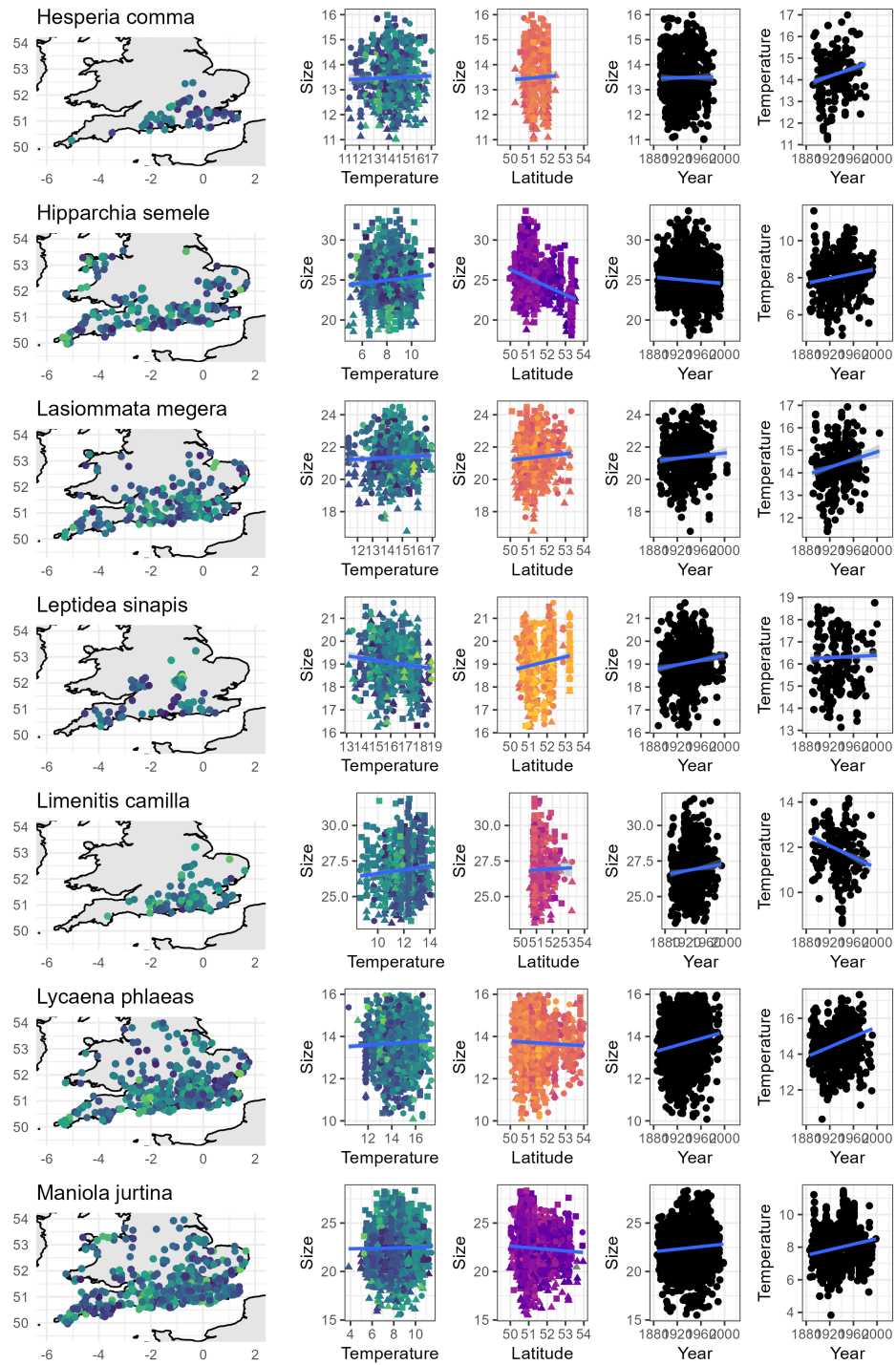
Figure S9: Repeat of Figure 4, excluding species coefficients where potential sex-specific response to predictors were identified. a) Year coefficient vs Size. Black line shows central estimate of regression without excluded species (slope: -0.0015, 95% interval: -0.0025 to -0.00055), red line shows main result with all species. b) Temperature Coefficient vs Red List Status. Boxplots do not include species with apparent sex-differences in responses. Estimated difference between Red listing status: -0.0029 (95% interval: -0.0051 to -0.0007).

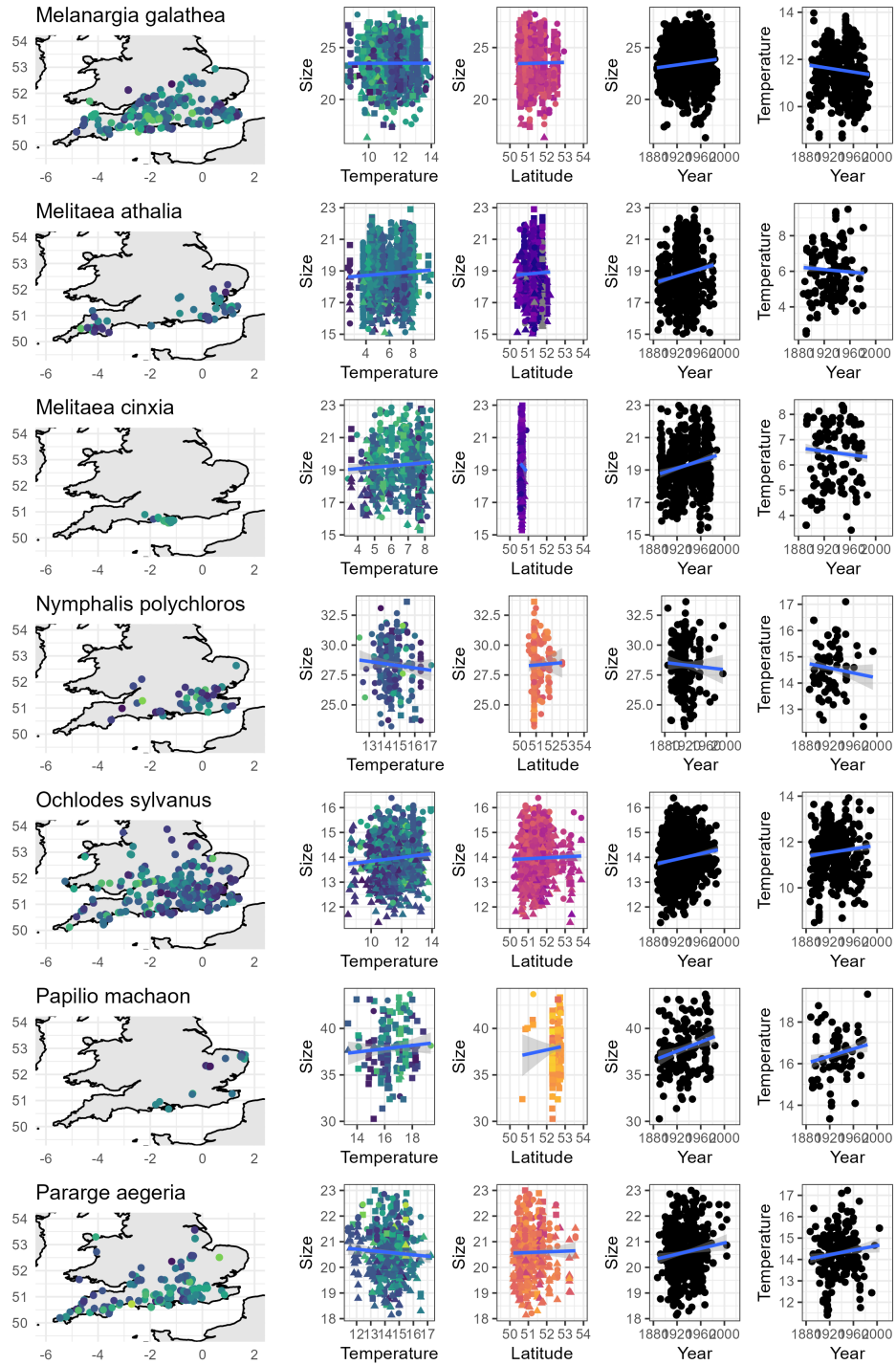
3 Specimen Spatio-temporal Distributions

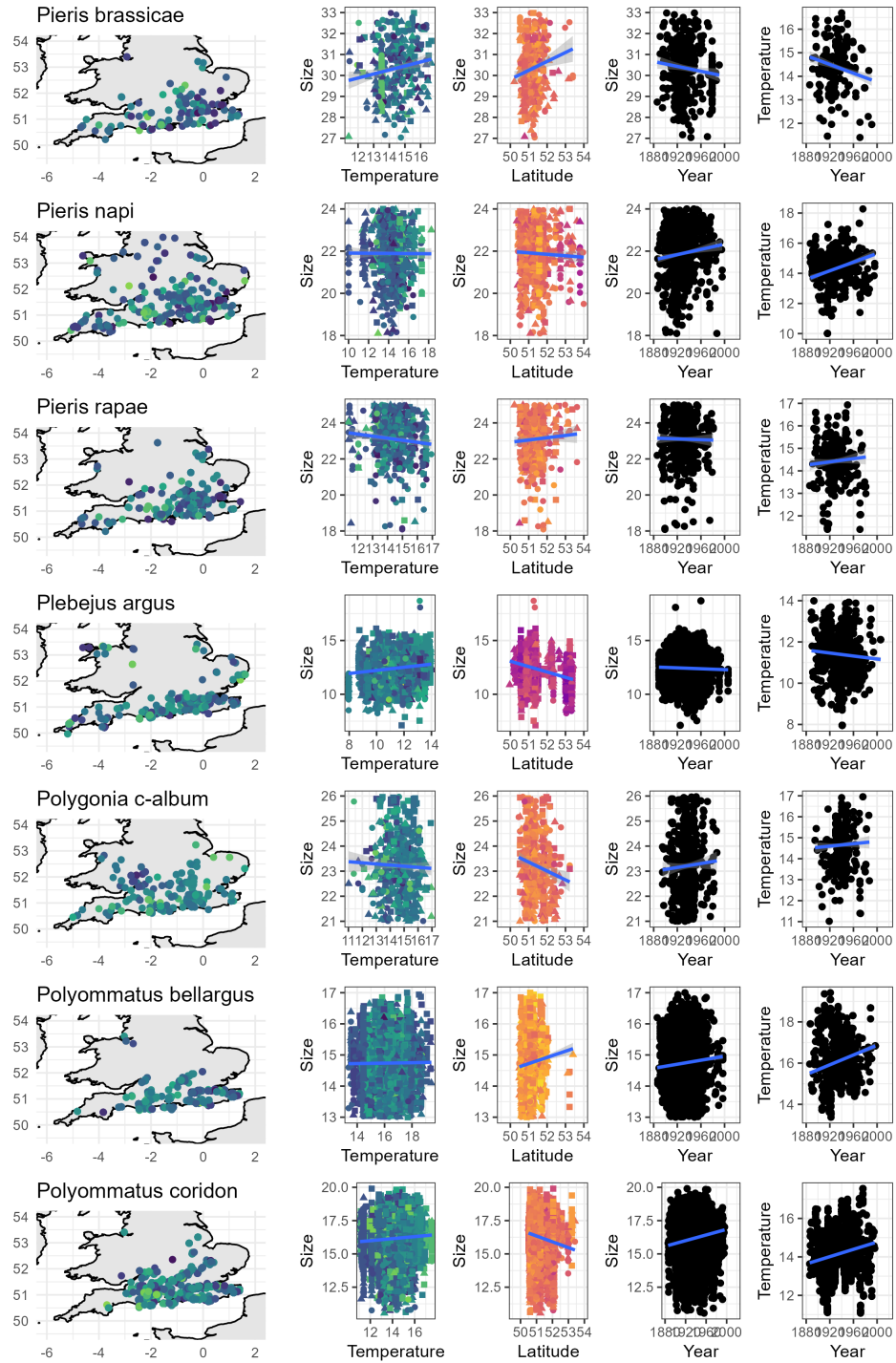


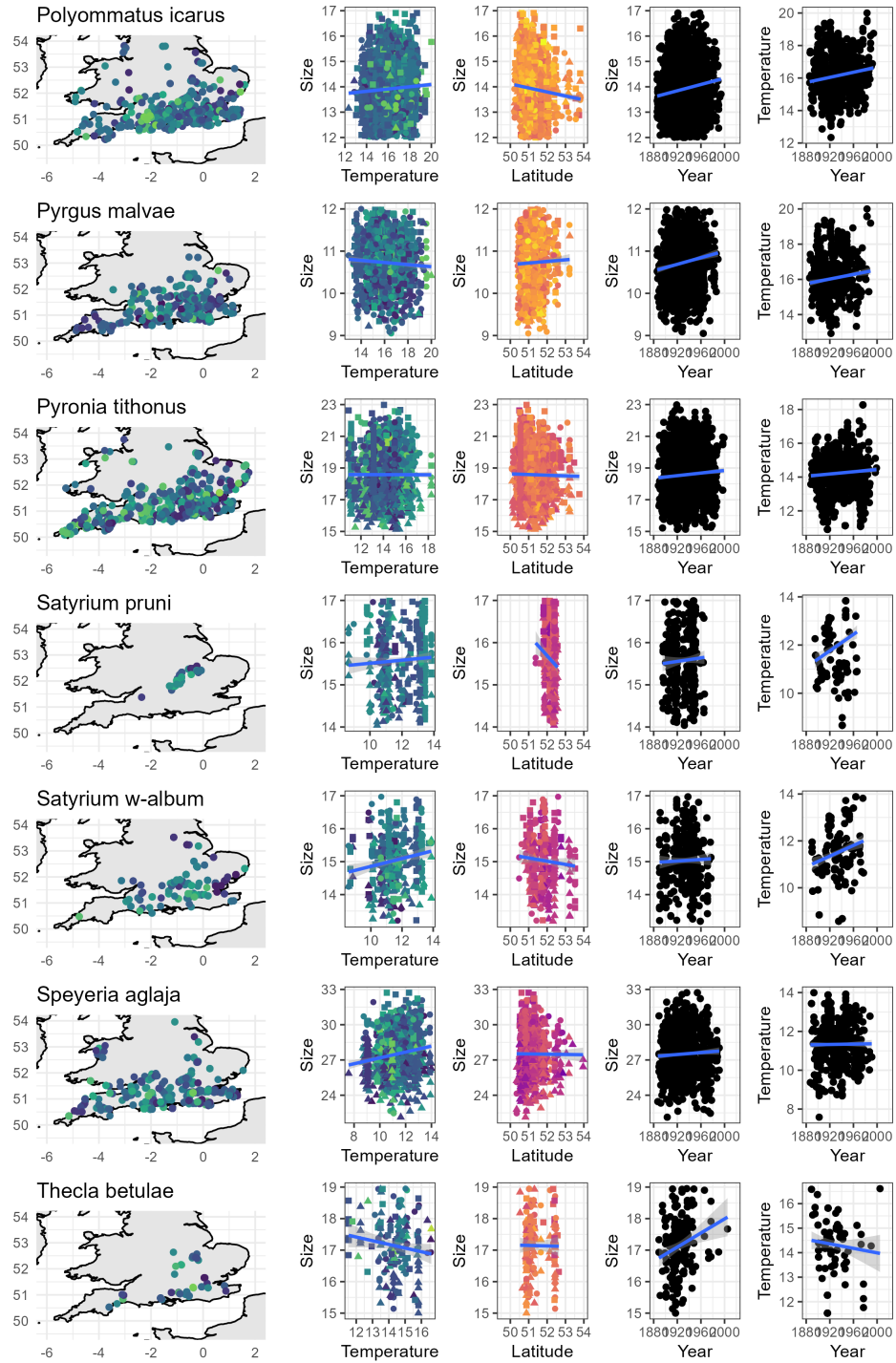












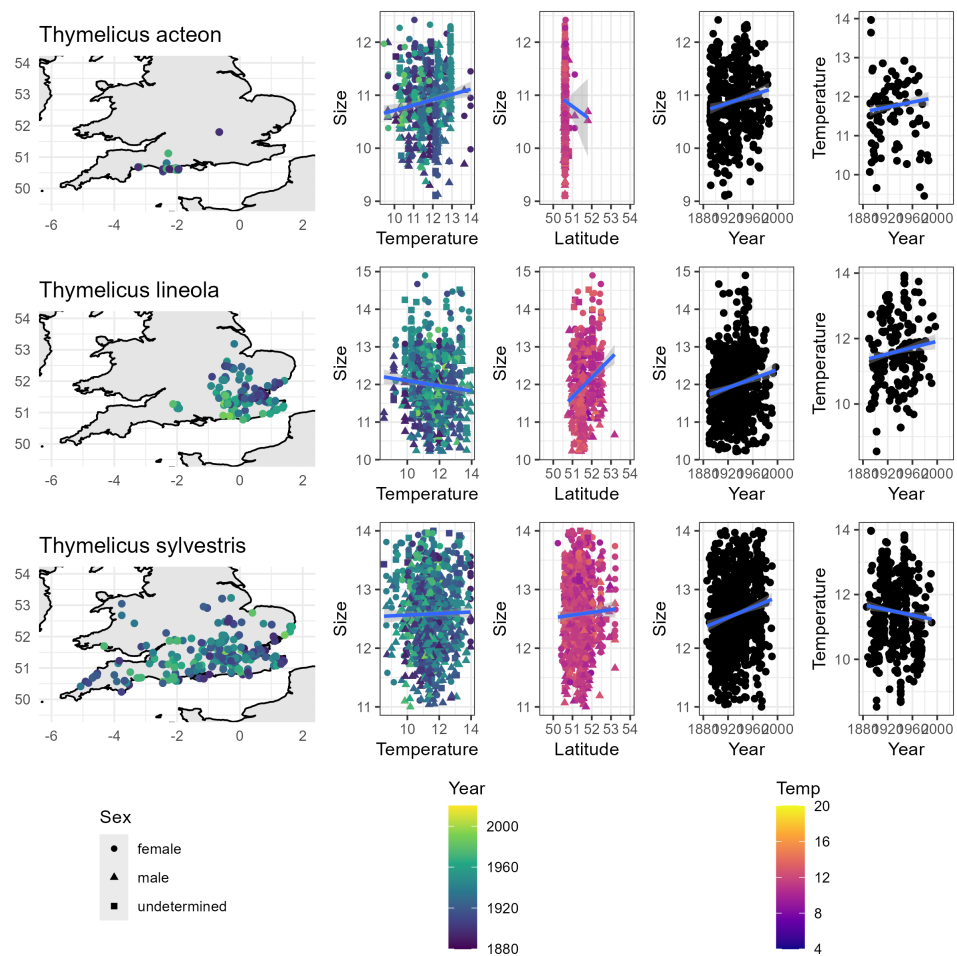
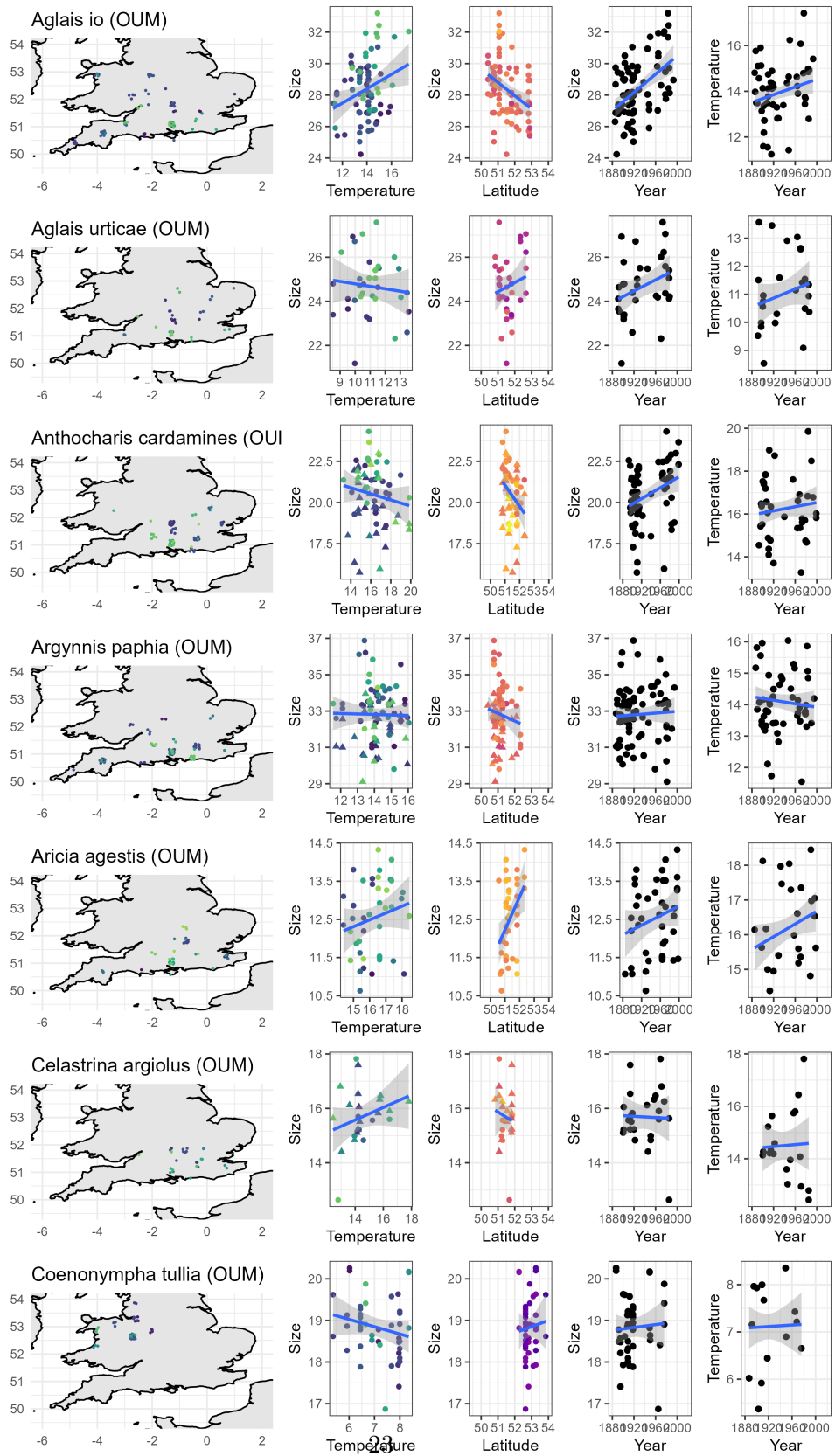
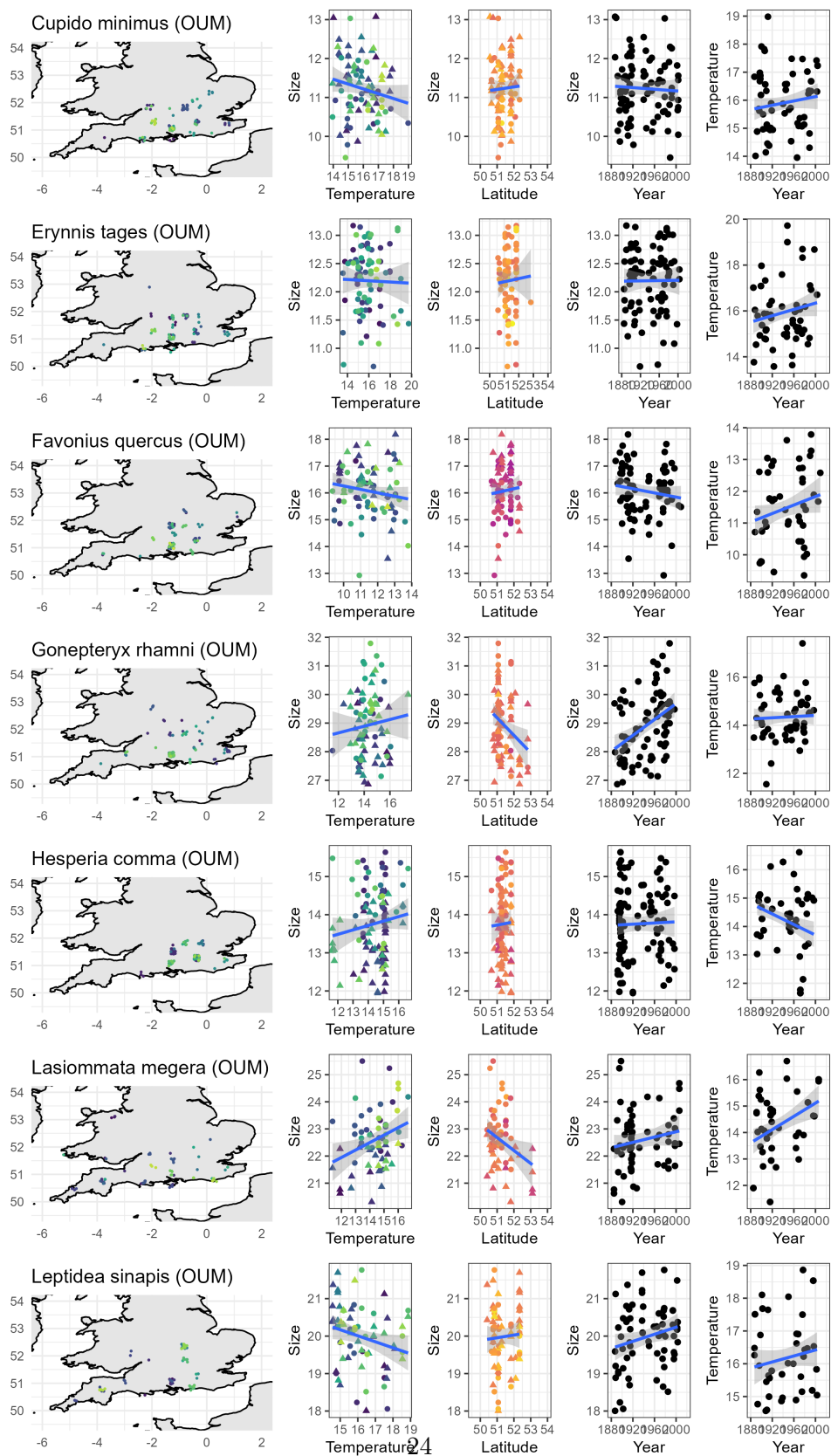
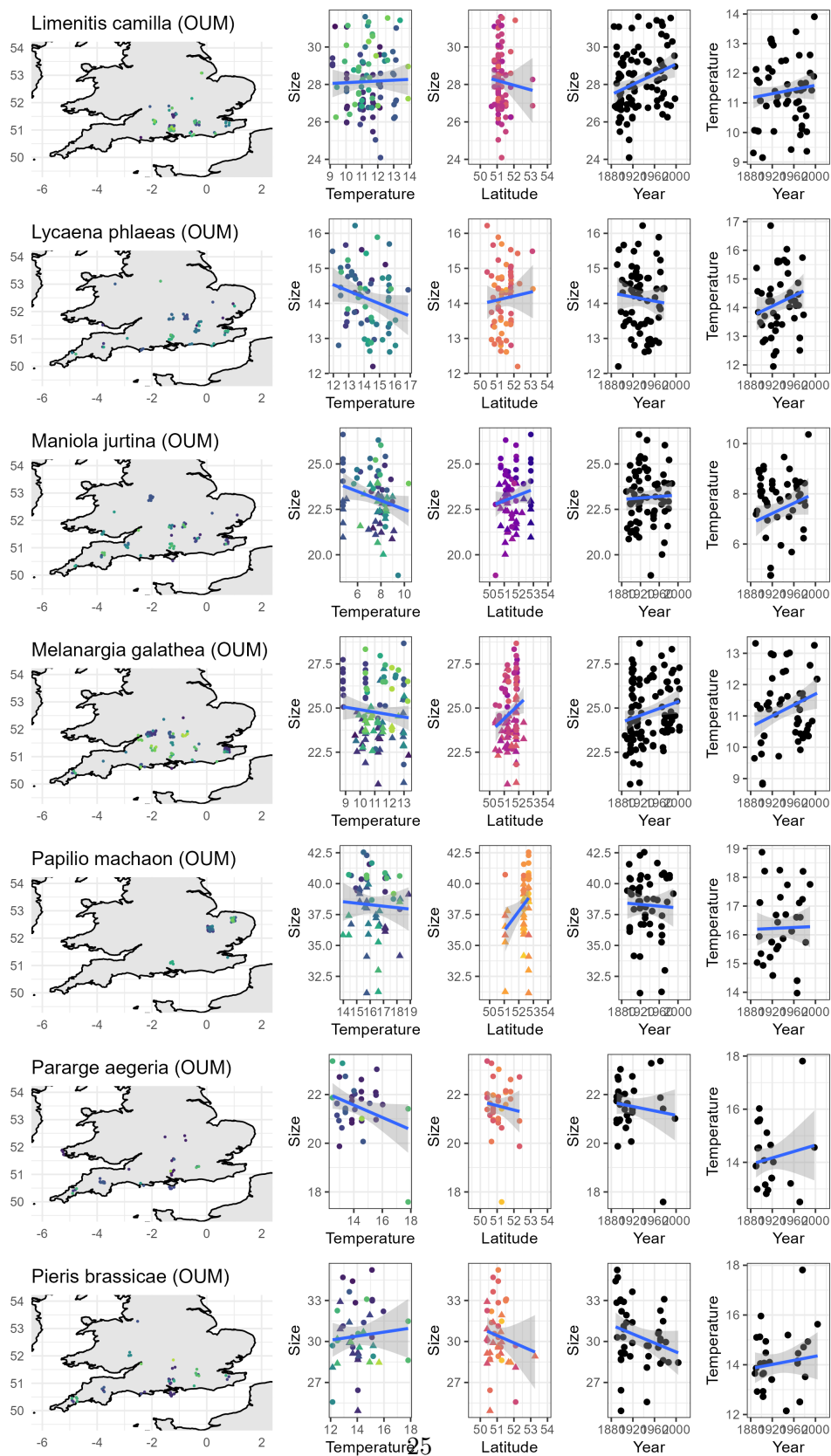


Figure S10: Distribution of records (post-filtering) from the NHM dataset for each species







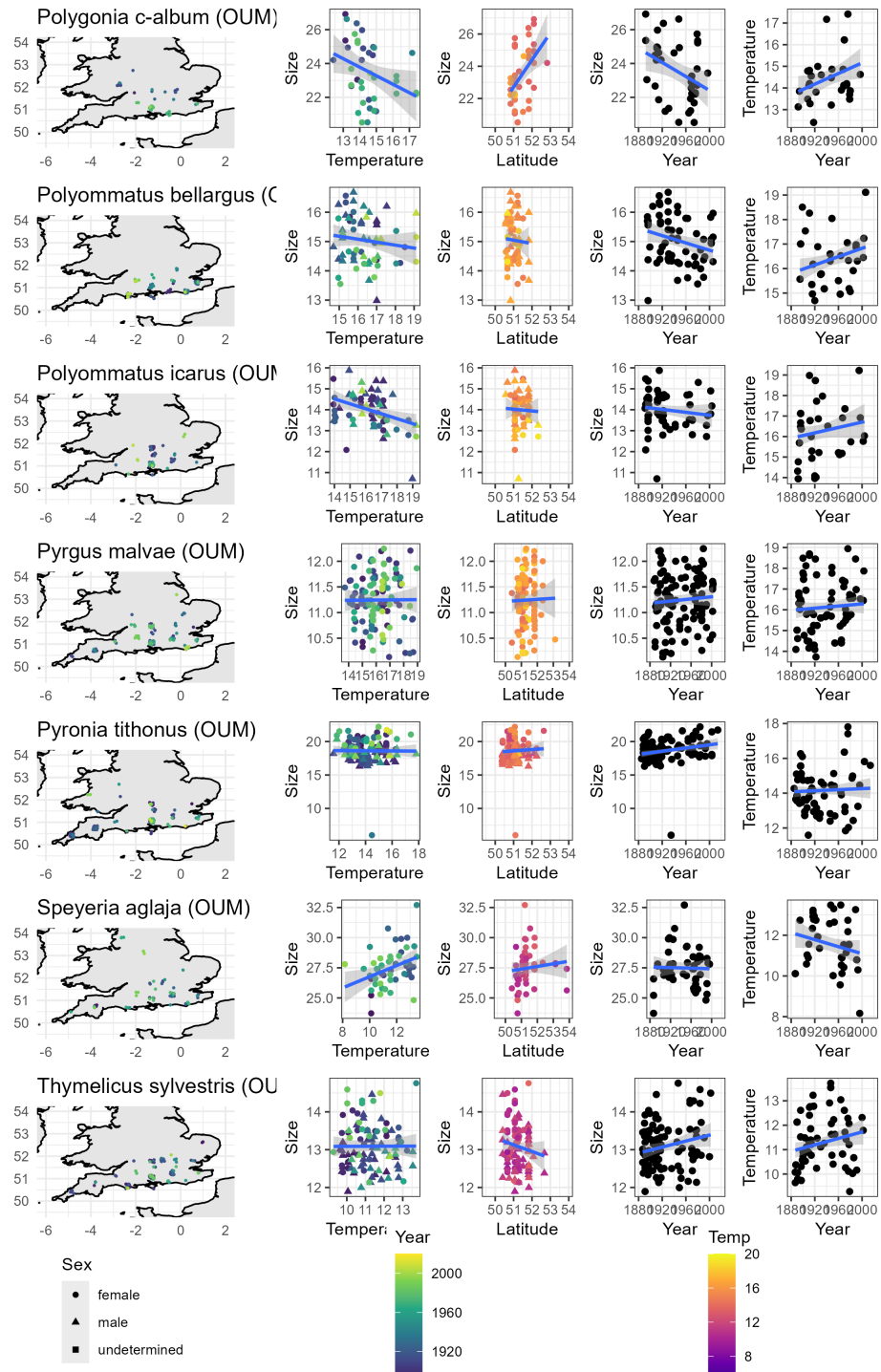


Figure S11: Distribution of records (post-filtering) from the NHM dataset for each species NB Locations on the map have been slightly jittered to reduce overlap.